Collective Intelligence for Architectural Programming in Mega Transportation Projects: Multi-Agent Simulation and Optimization

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ABSTRACT

Mega transportation projects (MTPs), as critical nodes in urban development, require architectural programming that balances multi-stakeholder interests and conflicting goals. Traditional rule-based methods struggle with dynamic environments and strategic interactions. This study proposes a collective intelligence-driven architectural programming system designed to be intelligent, collaborative, and resilient. First, a heterogeneous multi-agent system (MAS) models the behavioral patterns of five key stakeholders: government, investors, designers, contractors, and the public. Second, a hybrid training mechanism combining Proximal Policy Optimization (PPO) and Counterfactual Regret Minimization (CFR) enhances policy adaptation and conflict resolution. Third, a three-layer reinforcement learning architecture supports staged goal decomposition and dynamic behavioral coordination. Using Shanghai East Railway Station as a case study, results demonstrate that the proposed method outperforms baseline models in cost-efficiency, schedule control, social acceptance, and environmental outcomes, with improved convergence and stability. This study provides a resilient methodological approach for the architectural sector under complexity and uncertainty.

INTRODUCTION

Mega transportation projects (MTPs) serve as fundamental infrastructures in urban development, contributing not only to enhancing mobility and optimizing traffic flows but also to reshaping urban spatial structure, driving economic growth, and fostering social sustainability (Li et al. 2024). As global urbanization accelerates and population growth continues, cities face increasingly severe challenges such as traffic congestion, unequal resource distribution, and environmental degradation (Grimm et al. 2008). Against this backdrop, MTPs have emerged as crucial instruments for enhancing urban resilience and promoting regional synergy (Javed et al. 2022). These projects not only alleviate transportation bottlenecks and improve interregional connectivity but also facilitate functional urban restructuring and industrial upgrading, thereby accelerating regional economic integration (Zhou et al. 2020).

Compared to conventional infrastructure projects, MTPs are characterized by higher complexity and uncertainty throughout their design and implementation. Typically requiring massive investments and extended construction periods, MTPs encompass the integrated development of multiple transportation modes, such as rail transit, aviation hubs, and expressways. These projects must simultaneously balance technical feasibility, environmental sustainability, and social equity (AlKheder et al. 2021). Furthermore, the decision-making process involves a diverse array of stakeholders—including government agencies, investors, design teams, contractors, operators, and end users—whose varying needs, constraints, and priorities often diverge, making coordination and consensus-building particularly challenging (Li et al. 2019). Consequently, enhancing MTPs' adaptive capacity in uncertain environments, optimizing stakeholder collaboration, and ensuring long-term project sustainability have become pressing issues in the fields of architecture and engineering.

Architectural programming represents the critical starting point in the lifecycle of MTPs, determining spatial configuration, functional distribution, resource allocation, and operational strategy, with far-reaching impacts on subsequent design, construction, and operation phases (Jin and Tu 2024). Its core objective is to integrate the diverse demands of stakeholders, optimize the spatial and resource configuration under constraints of functionality, safety, and cost-efficiency, and enhance the project's ability to withstand future uncertainties (Hershberger 2015). However, architectural programming for MTPs faces multiple challenges, such as conflicts among multiple stakeholders (Tu and Jin 2024), high environmental uncertainty (Ansari 2019), and the complexity of multi-objective optimization (Lee et al. 2022). Therefore, architectural programming for MTPs urgently requires an intelligent, data-driven methodology to enhance collaborative efficiency, streamline decision-making, and improve the system's adaptability to uncertainty.

Collective intelligence provides a novel theoretical paradigm for architectural programming in MTPs. It refers to the emergence of superior decision-making and optimization capabilities through the interaction and learning of multiple agents, surpassing the abilities of individuals (Malone and Bernstein 2015). Its value lies in leveraging distributed computing and reinforcement learning to enhance adaptability and coordination in complex systems (Woolley et al. 2010). While collective intelligence has already demonstrated significant potential in areas such as autonomous driving, intelligent traffic scheduling, energy management, and robotic collaboration (Pu et al. 2022), its application in architecture and urban planning remains in an exploratory stage. Nevertheless, emerging studies have begun incorporating multi-agent systems (MAS) and reinforcement learning in spatial optimization, architectural design, and construction automation to enhance decision intelligence (Jin 2025).

Accordingly, this study proposes a collective intelligence-driven architectural programming method for MTPs based on multi-agent reinforcement learning (MARL), aiming to address the challenges of multi-stakeholder collaboration, dynamic optimization, and uncertainty adaptation. This study is structured around the following key questions:

RQ1: How can the key stakeholders in MTP architectural programming be systematically identified and modeled?

RQ2: How can MARL mechanisms facilitate collaborative optimization and collective decision-making?

RQ3: How can the proposed method's optimization performance, adaptability, and system resilience be validated within a practical project?

METHODOLOGY

The collective intelligence-based architectural programming framework in this study is illustrated in Figure 1. First, a MAS is constructed to perform domain modeling of heterogeneous stakeholders. Second, Proximal Policy Optimization (PPO) and Counterfactual Regret Minimization (CFR) are employed to address complex dynamic game problems. Third, a Hierarchical Reinforcement Learning (HRL) architecture is established to enable global policy coordination and iterative optimization feedback.

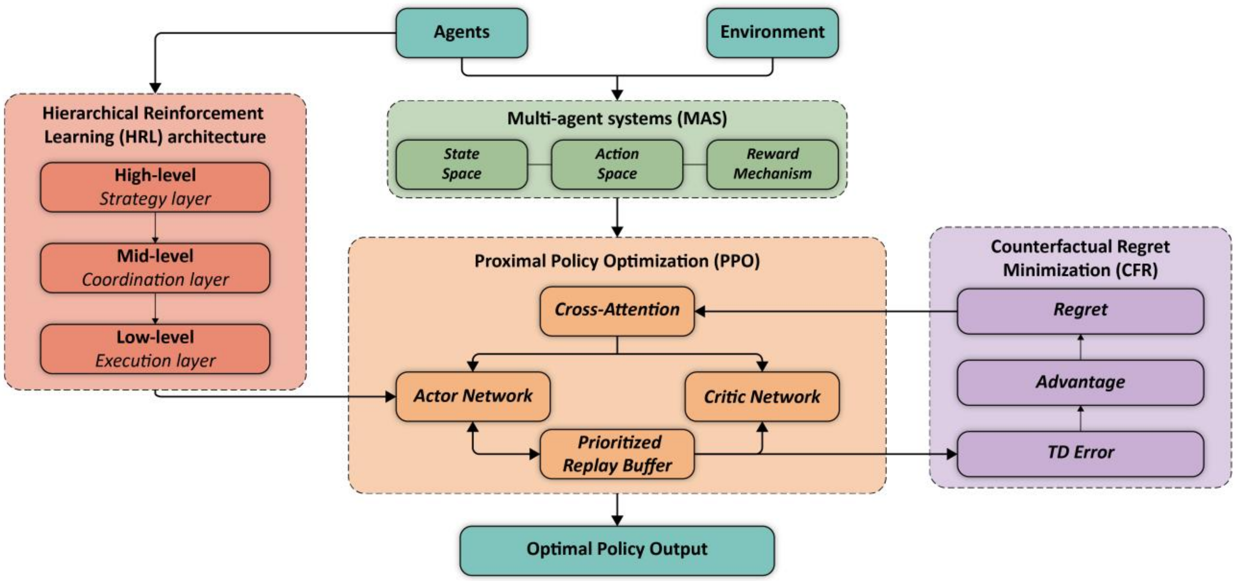

Figure 1. The collective intelligence-based architectural programming framework.

Multi-Agent Modelling for Stakeholder Representation. During the architectural programming of MTPs, the decision-making process exhibits multi-objective conflict. To address this complexity, this study constructs a heterogeneous agent model tailored to architectural programming scenarios based on the standard Multi-agent Markov Decision Process (MMDP) framework (Cruciol et al. 2013).

Let the agent set be denoted as $A = \{a_1, a_2, \ldots, a_N\}$, where N denotes the total number of agents. The entire programming system is represented by the quintuple $[S, A, P, R, \gamma]$: (1) S denotes the state space, which characterizes the global project state at time t, i.e., $S_t = \{s_1^t, s_2^t, \ldots, s_m^t\}$, where m is the dimensionality of the state space. Each agent can only observe a local subset of the global state, denoted as $S_i^t \subseteq S^t$. (2) A denotes the action space. At time t, each agent a_i executes an action $a_i^t \in A_i$, involving concrete programming behaviors such as policy adjustment, investment expansion or reduction, design iteration, and construction scheduling. (3) $P(S_{t+1} \mid S_t, A_t)$ is the state transition function, which describes the influence of the joint action $A_t = \{a_1^t, a_2^t, \ldots, a_N^t\}$ on the system evolution under the current state. (4) $r_i^t = R_i(S_t, A_t)$ is the instantaneous reward function for each agent, measuring the performance feedback of its current behavior under the system state. (5) $\gamma \in (0,1]$ is the discount factor, reflecting the importance of future returns. The objective of each agent's policy $\pi_i(a_i^t \mid s_i^t)$ is to maximize the cumulative return in the game, as shown in Equation (1):

$$J = \sum_{i=1}^{N} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t r_i^t\right] \tag{1}$$

Proximal Policy Optimization and Regret Minimization. To achieve policy optimization in dynamic environments, this study adopts PPO as the core learning algorithm and integrates the CFR mechanism to address the conflict between individual profit-seeking behavior and collective coordination in MTPs (Ponsen et al. 2011). A Centralized Training and Decentralized

Execution (CTDE) paradigm is employed to ensure that agents achieve global coordination capabilities under local observations.

At the algorithmic level, this study follows the standard PPO framework and constrains the magnitude of policy updates through a clipped surrogate objective function to ensure training convergence and robustness. The core optimization objective is expressed in Equation (2):

$$L(\theta) = \mathbb{E}_t[min(r_t(\theta)A_t, clip(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)] \tag{2}$$

where $r_t(\theta) = \frac{\pi_\theta(a^t|s^t)}{\pi_{\theta_{old}}(a^t|s^t)}$ is the probability ratio between new and old policies, A_t is the advantage function, and ϵ is the clipping threshold. However, pure PPO is prone to convergence to local optima in multi-objective games, leading to social-dilemma situations analogous to the "Prisoner's Dilemma." To overcome this limitation, this study introduces and refines a CFR-based policy correction module. Through counterfactual reasoning, the system computes the regret value after each decision round, defined as the payoff difference between the best unchosen action and the actually executed action, as shown in Equation (3):

$$R_i^t = \max_{a_i'} Q_i(s_i^t, a_i') - Q_i(s_i^t, a_i^t) \tag{3}$$

This hybrid mechanism dynamically adjusts the sampling probability of the policy distribution using the regret signal R_i^t, thereby guiding stakeholder agents to transition from competitive behavior toward cooperative behavior and gradually converge to a globally optimal policy equilibrium.

Hierarchical Reinforcement Learning Architecture. Based on the practical managerial hierarchy of architectural programming, this study further constructs a three-layer nested HRL architecture consisting of "strategy–coordination–execution" layers to achieve adaptive decoupling and global coordination of multi-level decision tasks in MTPs (Pateria et al. 2021).

First, the strategy layer simulates the decision logic of governments and core investors and is responsible for the generation of long-term goals. Agents define abstract sub-goals g based on policy and financial states. The high-level policy π_H is formulated as shown in Equation (4) and aims to maximize the long-term strategic return R_H^t:

$$\pi_H(s_H^t) = arg\max_\pi \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_H^t | s_H^0\right] \tag{4}$$

Second, the coordination layer serves as the central linkage hub of the architecture and is responsible for translating the high-level goals g into resource allocation and design trade-off directives. This layer is modeled as a Semi-Markov Decision Process (SMDP). The mid-level policy π_M is required to optimize the mid-term coordination return R_M^t while satisfying high-level constraints, as shown in Equation (5):

$$\pi_M(s_M^t | g) = arg\max_\pi \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_M^t | s_H^0, g\right] \tag{5}$$

Finally, the execution layer simulates the operational behaviors of designers and contractors and focuses on efficiency and compliance within short time scales. The low-level policy π_L depends on the current state and the task constraints issued by the mid-level layer, as shown in Equation (6):

$$\pi_L(a_L^t|s_L^t, g) = arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_L^t | s_L^0, g\right] \tag{6}$$

CASE STUDY

This study selects Shanghai East Railway Station as the empirical case to evaluate the effectiveness of the proposed framework. The case exhibits three critical characteristics: First, it involves a diverse array of stakeholders, forming a highly heterogeneous decision-making network. Second, the project has a long duration and densely packed milestones, and is influenced by a dynamic interplay of policy, capital, and societal factors. Third, the project was officially launched in 2023, providing a rich dataset and strong verifiability. The overall implementation workflow of the case study is illustrated in Figure 2.

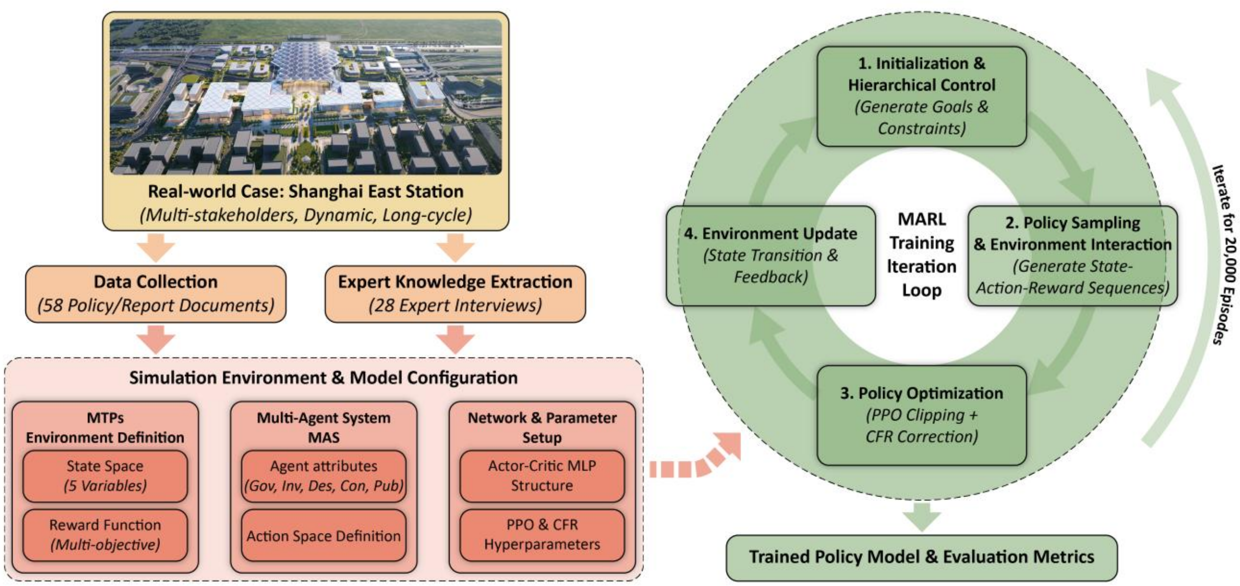

Figure 2. The workflow of the case study.

Simulation Environment Design. The simulation environment is constructed based on textual semantic analysis and expert interviews. This study first applied text mining techniques to process 58 policy documents, environmental reports, and media sources, extracting key attribute features of the project. In parallel, 28 senior industry experts were interviewed through semi-structured interviews to identify the core stakeholders and to clarify the causal logic of state transitions.

This study adopts a combination of Min–Max Normalization and an expert-threshold truncation method to determine the value ranges of the state variables (Table 1). First, extreme value intervals of each variable were identified based on documentary data. Second, project-specific feasibility thresholds were delineated according to expert consensus to remove theoretically possible but practically unacceptable intervals in real-world decision-making. Finally, the truncated decision sub-intervals were mapped into the standardized $[0, 1]$ space.

Table 1. State Variable Definitions.

Variable	Description	Range
Economic Feasibility	Reflects ROI, payback period, and capital metrics	[0.3, 0.8]
Social Acceptance	Captures public satisfaction and media sentiment	[0.2, 0.9]
Policy Compliance	Conformity with programming regulations and codes	[0.7, 1.0]
Construction Progress	Degree of project completion	[0.0, 0.3]
Environmental Impact	Includes carbon intensity and ecological burden	[0.4, 0.9]

Based on the expert interview results, the agent set is abstracted as: government (a_g), investor (a_i), designer (a_d), contractor (a_c), and public (a_p). The action space of each agent is defined according to the specific task categories in architectural programming (Table 2).

Table 2. Action Space Configuration.

Agent	Action Options	Primary State Impacted
a_g	Subsidy adjustment, regulation issuance	Policy compliance, social acceptance
a_i	Capital allocation, risk redistribution	Economic feasibility
a_d	Spatial optimization, green tech adoption	Environmental impact, policy compliance
a_c	Workflow optimization, method selection	Construction progress, environment
a_p	Feedback submission, complaint filing	Social acceptance

The reward functions are designed with primary objective terms and secondary incentive terms according to the goals of different stakeholders, jointly capturing the tension between local optimality and system-level coordination. All reward terms are normalized to the interval $[-1, 1]$, where positive values indicate improved goal attainment, while negative values indicate deviations from expected performance (Table 3).

Table 3. Reward Function Design.

Agent	Primary Reward Expression	Example Formula
a_g	Policy compliance + social benefits	$R_g = w_1 \cdot P + w_2 \cdot S$
a_i	ROI maximization − investment volatility	$R_i = w_1 \cdot E - w_2 \cdot \sigma_E$
a_d	Design efficiency − environmental burden	$R_d = w_1 \cdot U - w_2 \cdot C$
a_c	Progress rate − energy waste	$R_c = w_1 \cdot G - w_2 \cdot R$
a_p	Satisfaction improvement rate	$R_p = \Delta S_t$

Network and Parameter Configuration. This study constructs multi-layer perceptron (MLP) policy networks based on the Actor–Critic architecture to accommodate the decision-making requirements of agents at different hierarchical levels within the HRL framework. Given that architectural programming tasks in MTPs exhibit a moderately sized but strongly nonlinear-coupled state space, the network structures of different agents are configured with one to two hidden layers according to their task complexity and input dimensionality (Table 4).

Table 4. Policy Network Structures and Agent Objectives.

Agent	MLP Structure	Input Dimension	Output Dimension	Reward Function Goals
a_g	2-layer	5	2	Maximize policy compliance and social performance
a_i	2-layer	4	3	Maximize ROI, minimize investment risk
a_d	2-layer	4	4	Maximize functional efficiency, allocation
a_c	2-layer	3	3	Minimize construction duration, optimize cost
a_p	1-layer	3	1	Maximize satisfaction

The hyperparameter settings for MARL are shown in Table 5. The learning rates are separately configured for the actor and critic networks to balance training dynamics. A discount factor γ of 0.95 reflects the medium-to-long-term impact of decisions in architectural programming. Generalized Advantage Estimation (GAE) is used to reduce variance and enhance sample efficiency. All networks use ReLU activations, with target networks synchronized every 10 iterations.

Table 5. Reinforcement Learning Hyperparameters.

Parameter	Value
Learning Rate (Actor/Critic)	1e-4 / 1e-3
Discount Factor (γ)	0.95
PPO Clip Range (ε)	0.2
Advantage Estimation	GAE (λ= 0.95)
CFR Learning Rate	Adaptive (initial: 0.1)
Batch Size	8192
Total Training Episodes	20,000
Network Sync Interval	Every 10 iterations
Optimizer	Adam

Model Training and Evaluation. The model training process encompasses a complete evolutionary cycle from initialization to policy iteration. In the initialization phase, the system initializes the state distribution based on expert knowledge and constructs the corresponding policy and value networks for the five categories of agents. In policy sampling phase, agents interact with the simulation environment according to their current policies, generating state–action–reward trajectories, while a prioritized experience replay mechanism is employed to enhance learning efficiency. In the policy optimization phase, PPO is responsible for improving policy returns within a trust-region–constrained optimization regime, while CFR dynamically computes regret signals to guide agents in identifying and correcting non-cooperative behaviors.

To test the robustness of the training outcomes, this study further conducts an ablation study with three comparative configurations: standard MAPPO, MAPPO + CFR, and MAPPO + HRL. These settings are designed to systematically decouple and validate the independent contributions of the CFR mechanism and the HRL architecture in resolving conflicts and supporting long-horizon programming. Finally, based on the real-time logging of policy mutual information (PMI), goal entropy, and coordination index in each iteration, the model is evaluated for convergence stability within 20,000 training episodes. These indicators are fed back as adaptive signals to dynamically adjust hyperparameters, ensuring that the final output policies represent a rigorously validated globally optimal solution.

RESULTS

As shown in Figure 3, the system's overall return increases rapidly during the initial 3,000 training episodes, indicating that agents can formulate effective preliminary strategies early in the exploration phase. Between 10,000 and 16,000 episodes, returns rise at an accelerated pace and begin to converge after 16,000 episodes. At the individual agent level, the government and investor agents exhibit steadily increasing returns, attributable to relatively stable objective structures. In contrast, the design and construction agents undergo mid-phase adjustments, reflecting their strategic balancing among different competing goals. The public agent, which is highly sensitive to systemic dynamics, demonstrates greater initial volatility but eventually stabilizes, indicating its adaptive responsiveness within the environment.

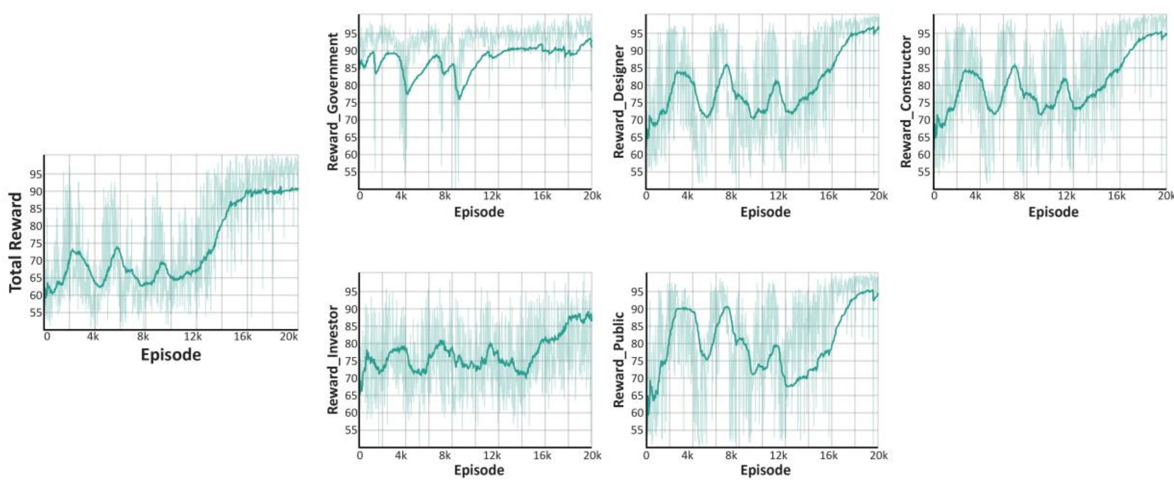

Figure 3. The total reward and agent-specific rewards.

As shown in Figure 4, the proposed framework outperforms all baseline models across all evaluation metrics: (1) Economic feasibility improves from 0.48 to 0.74, surpassing MAPPO by 0.08. This suggests that agents have reached more optimal ROI strategies through iterative game-based learning. (2) Social acceptance rises from 0.35 to 0.71, attributed to the incorporation of public preferences into the coordination layer via the strategic layer, thereby enhancing the system's adaptability to social feedback. (3) Policy compliance reaches a maximum of 0.91, indicating that the integration of HRL significantly enhances alignment between agent behaviors and regulatory frameworks.

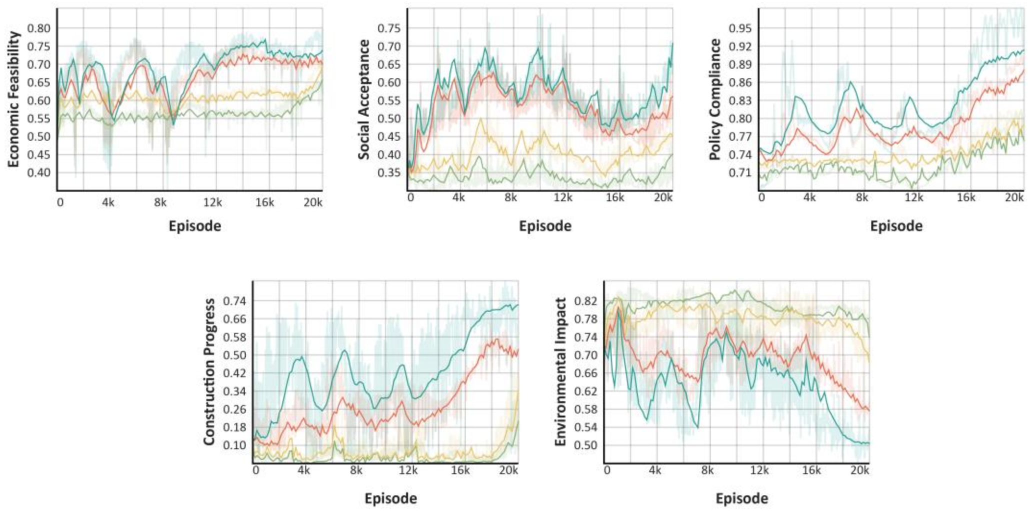

Figure 4. Comparative analysis of key decision metrics.

As illustrated in Figure 5, starting from 5,000 episodes, PMI rises rapidly, reaching 0.31 by 20,000 episodes, reflecting enhanced coupling among agent strategies and a transition from competitive to cooperative dynamics. Goal distribution entropy decreases from an initial value above 0.75 to approximately 0.65, indicating a shift from dispersed to more focused high-level goal generation. While maintaining goal diversity, the system exhibits improved strategic convergence. The coordination index increases significantly from 15.2% to 68%, signifying a marked improvement in the coherence of joint actions.

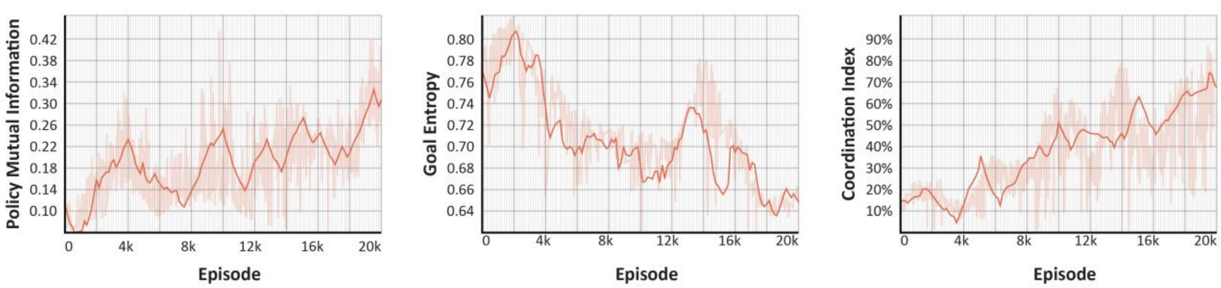

Figure 5. Analysis of key behavioral coordination metrics.

DISCUSSIONS AND CONCLUSIONS

Targeting the inherent characteristics of MTPs, namely multi-stakeholder collaboration, high-feedback uncertainty, and nonlinear system evolution, this study develops an architectural programming methodology that integrates MAS, PPO, and HRL. Using Shanghai East Railway Station as a case study, the proposed framework demonstrates robust effectiveness and generalizability across multiple performance dimensions.

The coordination mechanism embedded within the collective intelligence framework is not merely a means to enforce consensus among stakeholders but serves as a structural absorber of strategic heterogeneity. The MAS model emphasizes behavioral diversity and localized perception, while the CTDE and CFR mechanisms ensure that individual agents can learn independently while gradually aligning their strategies through mutual information exchange and feedback regulation (Ponsen et al. 2011). In the context of architectural programming, where inter-role interaction and consensus generation are critical, this mechanism more authentically simulates the evolution of objectives and the resolution of conflicts. In other words, coordination is not treated as a predefined constraint but as an outcome of learning, rendering the process both interpretable and adaptive.

Moreover, the proposed method can be interpreted as a mechanism for constructing organizational resilience in the context of complex infrastructure development. Enabled by the rolling feedback architecture of the HRL-based hierarchical strategy system, the model continuously reconstructs its strategic distribution in response to shifting goals, resource perturbations, and interest conflicts, thereby exhibiting a form of systemic self-healing (Veloso and Krishnamurti 2023). Notably, the observed phenomenon of “path restructuring” during policy evolution indicates not only an ability to absorb short-term disturbances but also reveals the structural evolution potential of the architectural programming system under multi-agent learning dynamics.

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REFERENCES

AlKheder, S., D. Talib, and A. Al-Mutairi. 2021. “Scenario-based preference modeling to examine the robustness of airport mega projects initiatives.” *Science of the Total Environment* 797.

Ansari, R. 2019. “Dynamic Simulation Model for Project Change-Management Policies: Engineering Project Case.” *Journal of Construction Engineering and Management* 145 (7).

Cruciol, L. L. B. V., A. C. de Arruda, W. G. Li, L. H. Li, and A. M. F. Crespo. 2013. “Reward functions for learning to control in air traffic flow management.” *Transportation Research Part C-Emerging Technologies* 35:141-155.

Grimm, N. B., S. H. Faeth, N. E. Golubiewski, C. L. Redman, J. G. Wu, X. M. Bai, and J. M. Briggs. 2008. “Global change and the ecology of cities.” *Science* 319 (5864):756-760.

Hershberger, R. 2015. *Architectural programming and predesign manager*: Routledge.

Javed, A. R., F. Shahzad, S. U. Rehman, Y. Bin Zikria, I. Razzak, Z. Jalil, and G. D. Xu. 2022. “Future smart cities requirements, emerging technologies, applications, challenges, and future aspects.” *Cities* 129.

Jin, S. T. 2025. “The Application of Collective Intelligence in the Construction Industry: A Review of the Current State, Challenges, And Opportunities.” *Architectural Engineering and Design Management*:1-26.

Jin, S. T., and H. J. Tu. 2024. "Current status and research progress in architectural programming: A comparative analysis between China and other countries." *Frontiers of Architectural Research* 14 (2):487-509.

Lee, Z. P., R. A. R. Abdul, and I. D. Shu. 2022. "Application of decision support tool in design-build projects: a quasi-experiment with novice decision makers." *Built Environment Project and Asset Management* 12 (4):537-555.

Li, L., H. Y. Luan, M. Q. Yuan, and R. Y. Zheng. 2024. "Knowledge fusion-driven sustainable decision-making for mega transportation infrastructure projects." *Engineering Construction and Architectural Management* ahead-of-print (ahead-of-print).

Li, Y. K., Y. L. Han, M. X. Luo, and Y. Zhang. 2019. "Impact of Megaproject Governance on Project Performance: Dynamic Governance of the Nanning Transportation Hub in China." *Journal of Management in Engineering* 35 (3).

Malone, T. W., and M. S. Bernstein. 2015. *Handbook of Collective Intelligence*: The MIT Press.

Pateria, S., B. Subagdja, A. H. Tan, and C. Quek. 2021. "Hierarchical Reinforcement Learning: A Comprehensive Survey." *ACM Computing Surveys* 54 (5).

Ponsen, M., S. de Jong, and M. Lanctot. 2011. "Computing Approximate Nash Equilibria and Robust Best-Responses Using Sampling." *Journal of Artificial Intelligence Research* 42:575-605.

Pu, Z. Q., J. Q. Yi, Z. Liu, T. H. Qiu, J. L. Sun, and F. M. Li. 2022. "Knowledge-based and Data-driven Integrating Methodologies for Collective Intelligence Decision Making: A Survey." *Acta Automatica Sinica* 48 (3):627-643.

Tu, H. J., and S. T. Jin. 2024. "A group decision-making model for architectural programming in megaprojects." *Engineering, Construction and Architectural Management* 31 (13):342-368.

Veloso, P., and R. Krishnamurti. 2023. "Spatial synthesis for architectural design as an interactive simulation with multiple agents." *Automation in Construction* 154.

Woolley, A. W., C. F. Chabris, A. Pentland, N. Hashmi, and T. W. Malone. 2010. "Evidence for a Collective Intelligence Factor in the Performance of Human Groups." *Science* 330 (6004):686-688.

Zhou, M., H. Y. Zhuang, and S. J. Fang. 2020. "Innovations at Beijing Daxing International: the world's biggest airport terminal." In *Proceedings of the Institution of Civil Engineers-Civil Engineering* 173 (3):113-118.